Characteristics of Images

- Electromagnetic energy may be detected either photographically or electronically.

- It is important to distinguish between the terms **images** and **photographs** in remote sensing.

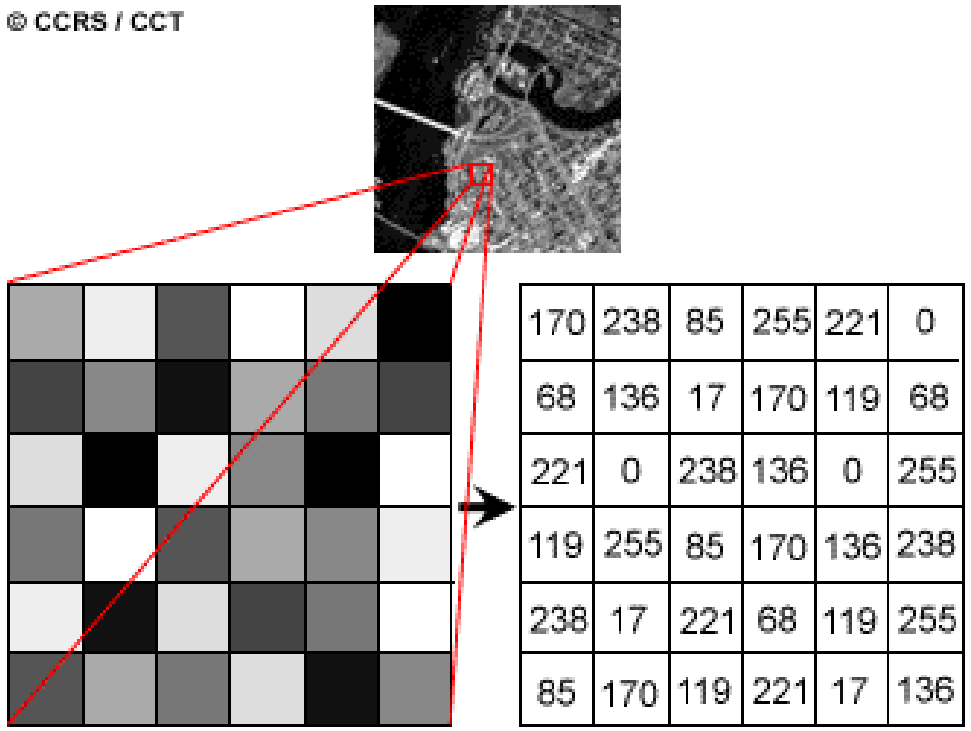
- The photographic process uses chemical reactions on the surface of light-sensitive film to detect and record energy variations.

- An image refers to any pictorial representation, regardless of what wavelengths or remote sensing device has been used to detect and record the electromagnetic energy.



- A **photograph** refers specifically to images that have been detected as well as recorded on photographic film.

- Photos are normally recorded over the wavelength range from 0.3 µm to 0.9 µm - the visible and reflected infrared.



- A photograph could also be represented and displayed in a **digital** format by subdividing the image into small equal-sized and shaped areas, called picture elements or **pixels**, and representing the brightness of each area with a numeric value or **digital number**.

- The photograph was scanned and subdivided into pixels with each pixel assigned a digital number representing its relative brightness.

- The computer displays each digital value as different brightness levels.

- Sensors that record electromagnetic energy, electronically record the energy as an array of numbers in digital format right from the start.

- These two different ways of representing and displaying remote sensing data, either pictorially or digitally, are interchangeable as they convey the same information (although some detail may be lost when converting back and forth).

- The information from a narrow wavelength range is gathered and stored in a **channel**, also sometimes referred to as a **band**.

- We can combine and display channels of information digitally using the three primary colours (blue, green, and red).

- The data from each channel is represented as one of the primary colours and, depending on the relative brightness (i.e. the digital value) of each pixel in each channel.

- When we use this method to display a single channel or range of wavelengths, we are actually displaying that channel through all three primary colours.





- Because the brightness level of each pixel is the same for each primary colour, they combine to form a **black and white image**, showing various shades of grey from black to white.

- When we display more than one channel each as a different primary colour, then the brightness levels may be different for each channel/primary colour combination and they will combine to form a **colour image**.